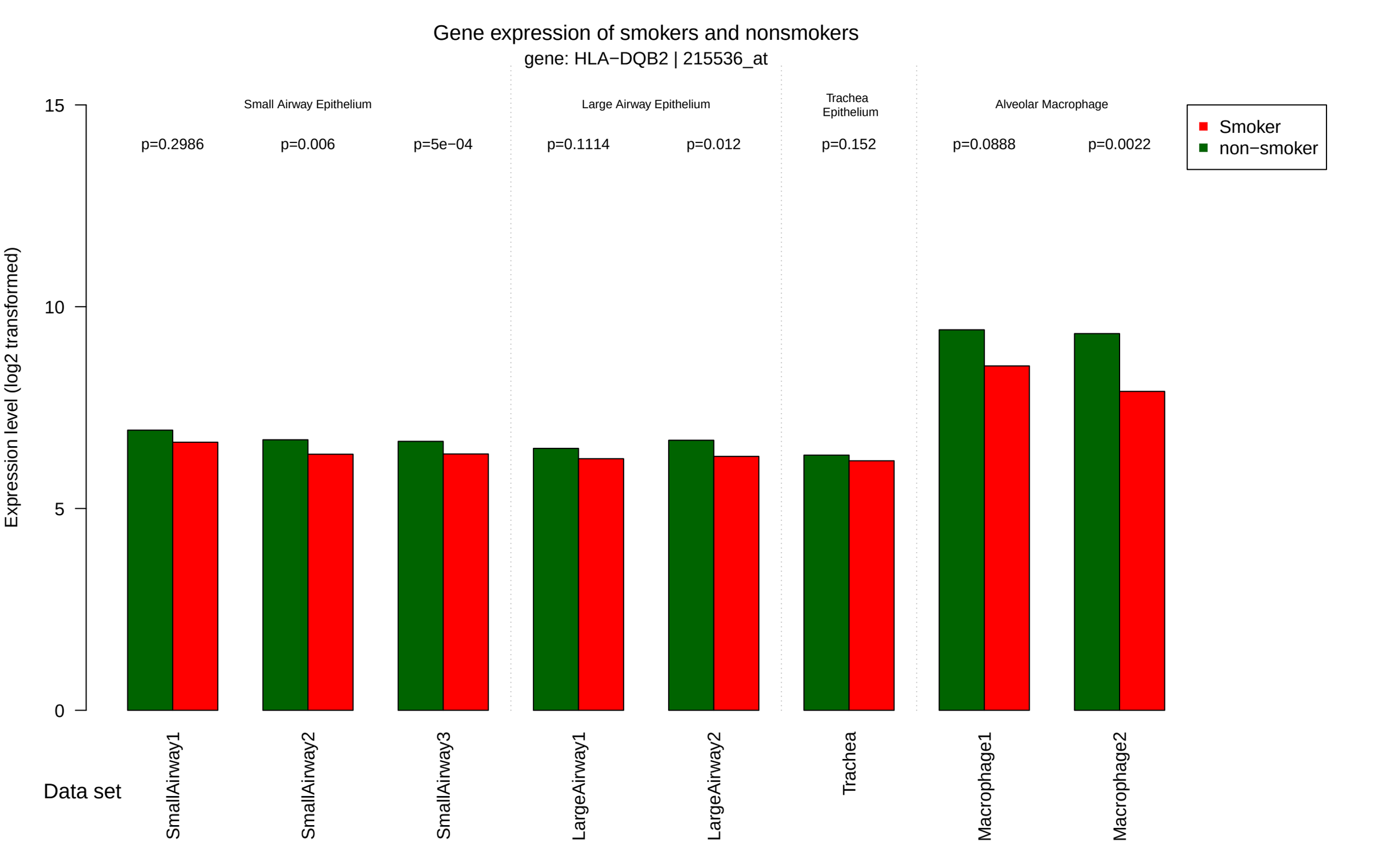
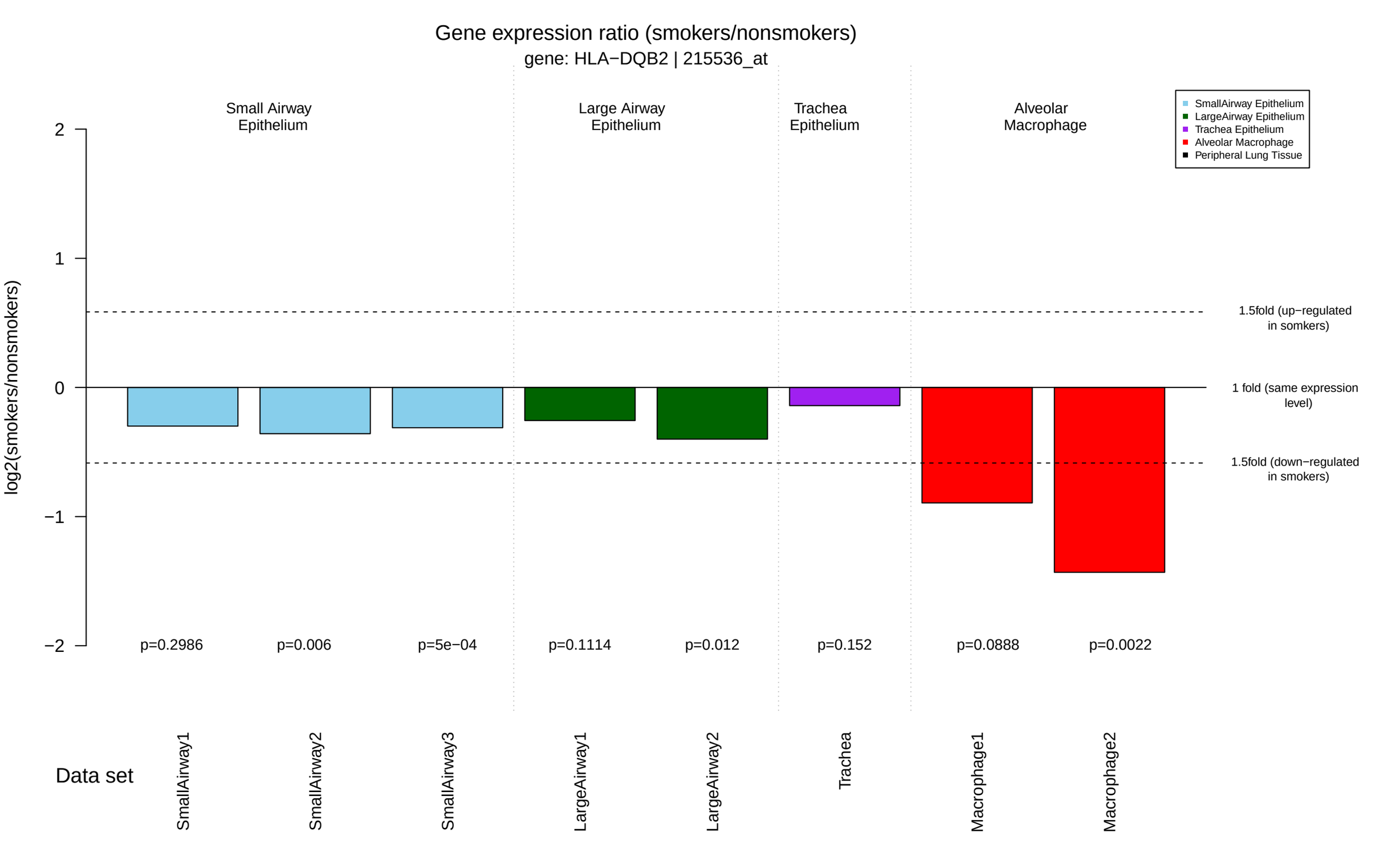
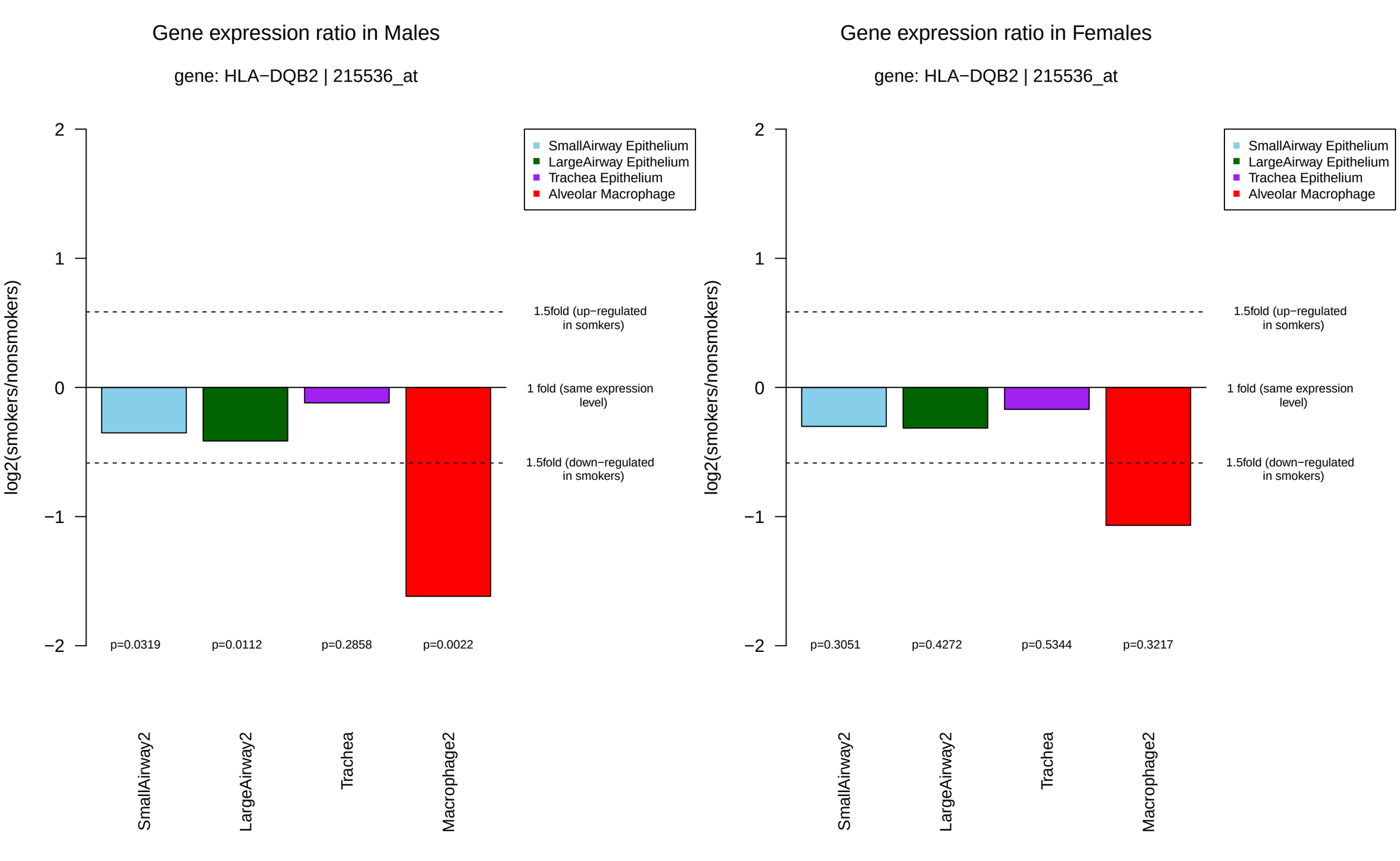
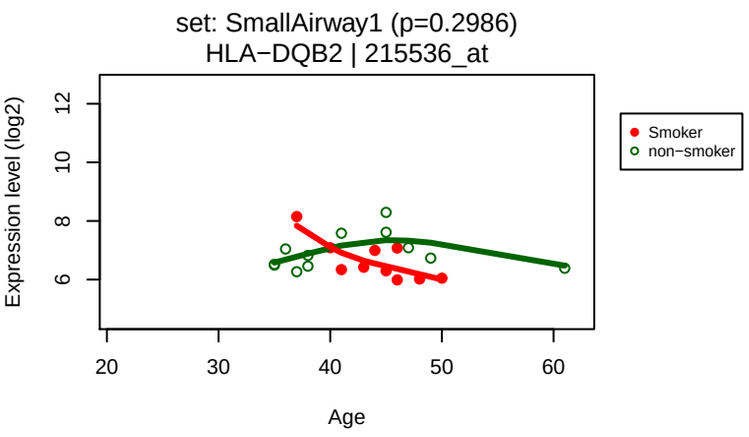
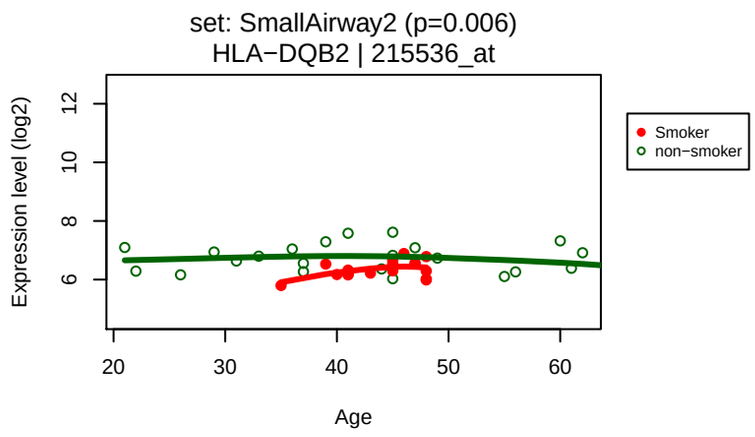
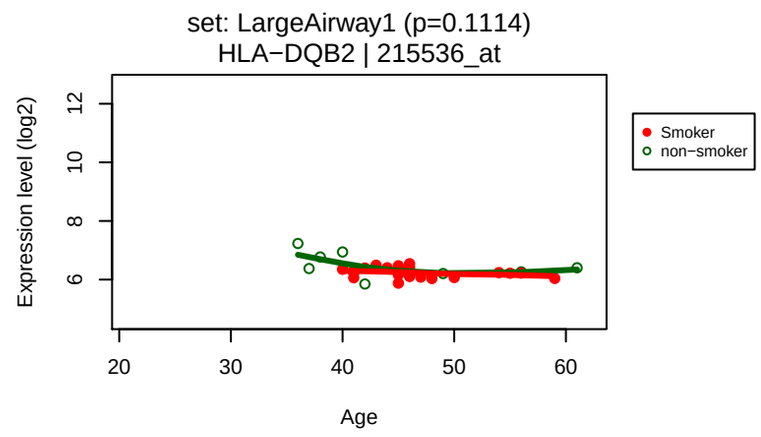

Gene expression of smokers and nonsmokers
gene: HLA−DQB2 | 215536_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2986
p=0.006
p=5e−04
p=0.1114
p=0.012
p=0.152
p=0.0888
p=0.0022
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

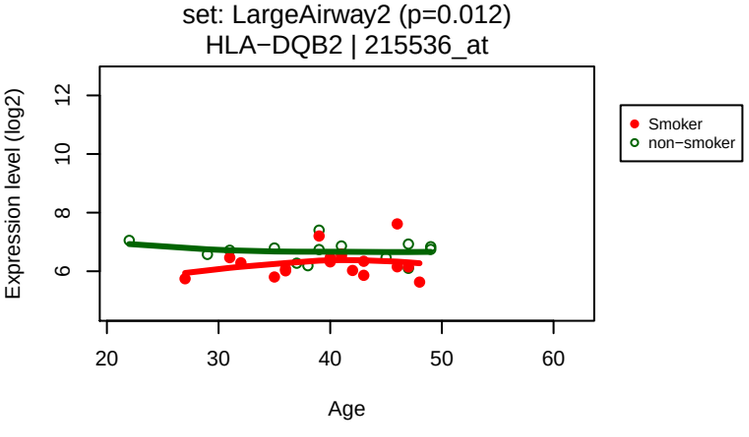

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQB2 | 215536_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2986
p=0.006
p=5e−04
p=0.1114
p=0.012
p=0.152
p=0.0888
p=0.0022
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

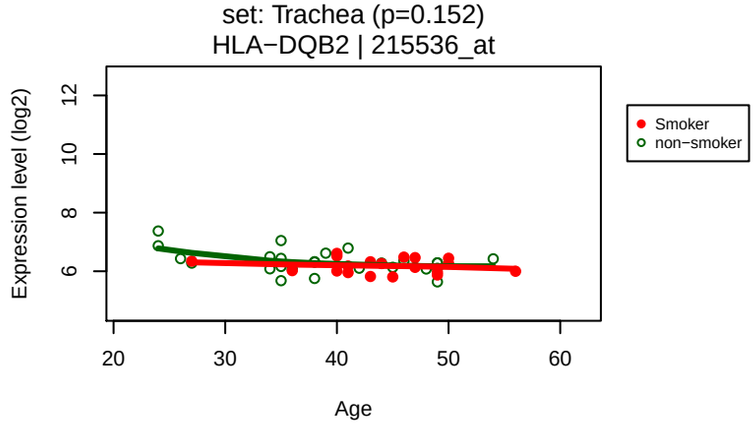

Gene expression ratio in Males
gene: HLA−DQB2 | 215536_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0319
p=0.0112
p=0.2858
p=0.0022
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQB2 | 215536_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3051
p=0.4272
p=0.5344
p=0.3217
SmallAirway2
LargeAirway2
Trachea
Macrophage2

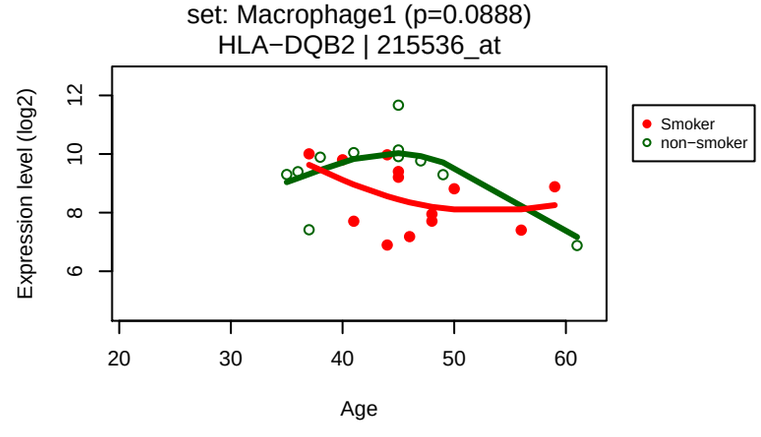

set: SmallAirway1 (p=0.2986)
HLA−DQB2 | 215536_at
Expression level (log2)
Age
Smoker
non−smoker

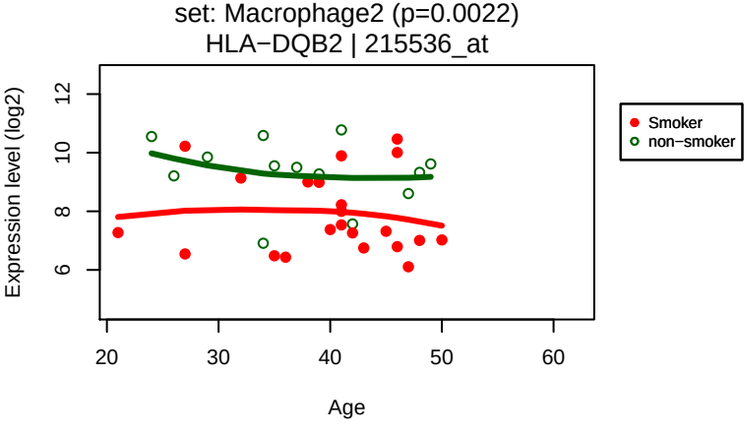

set: SmallAirway2 (p=0.006)
HLA−DQB2 | 215536_at
Expression level (log2)
Age
Smoker
non−smoker

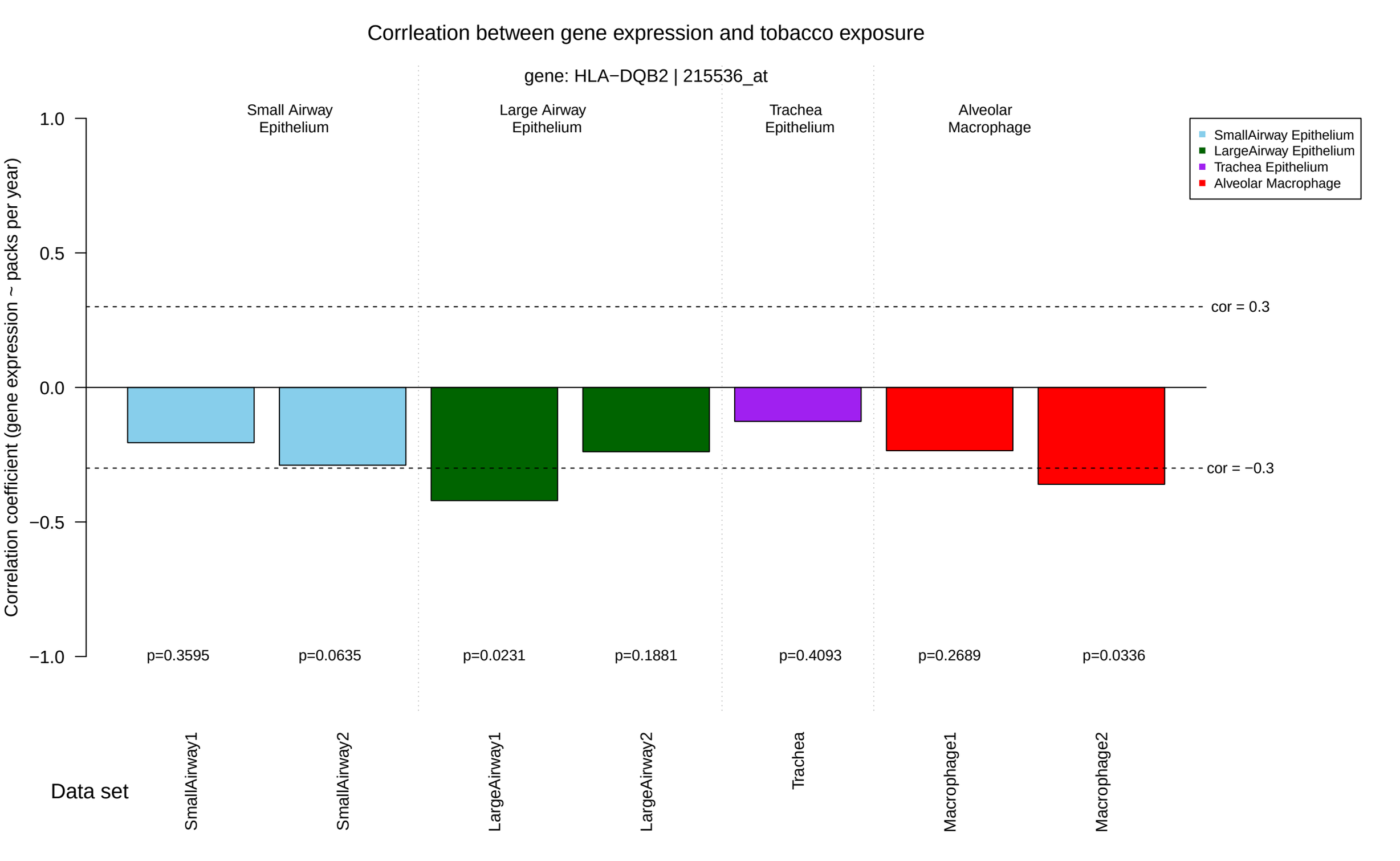

set: LargeAirway1 (p=0.1114)
HLA−DQB2 | 215536_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.012)
HLA−DQB2 | 215536_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.152)
HLA−DQB2 | 215536_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0888)
HLA−DQB2 | 215536_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0022)
HLA−DQB2 | 215536_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQB2 | 215536_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3595
p=0.0635
p=0.0231
p=0.1881
p=0.4093
p=0.2689
p=0.0336
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2